

Education Big Fish datasets — primer for researchers

Last synced: 2026-09-22

Audience: Charles (and Alex) — dissertation HERO / assortativity work, no NCES survey background required.

Purpose: Enough context to read Alex’s **NELS:88** and **HS&B:80** Big Fish panels, run porch screens, and talk honestly about **assortativity**, **peer context**, and **what education can (and cannot) test** in our model.

Standalone: One document; paths below are repo-relative from project root.

PDF: Charles converts locally: `./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/re_entry/HEROs_and_PASSes/EDUCATION_dataset_primer.md`

1. What these datasets are (plain English)

Both files are **Alex-built “Big Fish” panels** on classic **U.S. national education longitudinal surveys**. They are **not** raw NCES microdata you have to construct from scratch — peer leave-one-out (LOO) means, unit sizes, and analytic flags are **already computed**.

Dataset	Full name	Cohort	Pond (unit)	Outcome horizon
NELS:88	National Education Longitudinal Study	1988 8th-grade cohort, peer snapshot at 10th grade	High school (10th-grade classmates in sample)	Degree attainment by 2000 (~age 26)
HS&B:80	High School & Beyond	1980 high-school Sophomore and Senior samples (separate)	School × cohort wave	Degree attainment by 1986 (~age 22–24)

Alex’s intent (Sep 2026): Education is **“gold for the community”** — public, familiar domain to show **peer ponds, sorting (assortativity)**, and **peer context vs long-run outcomes** without sports framing.

Repo paths:

Panel CSV
<code>datasets/nels88/nels88_big_fish_panel.csv</code>
<code>datasets/hsb80/hsb80_big_fish_panel.csv</code>

Status in repo: CSVs exist; no `education_sandbox/` or 3×3 mosaic yet (unlike Army, MBB, football). Screening = next build step.

2. One row = what?

Each row is **one sampled student** in the analytic file.

- **Not** a full school census — only students **in the NCES sample** for that school appear as peers.
- `peer_n` ≈ 13 and `unit_n` ≈ 14 (NELS, min-10 sample) means roughly **14 sampled kids from that school**, not “everyone in the building.”
- Peer LOO is: *mean test score of the **other sampled students** in my school cell*, excluding self.

Time structure: Peer context is a **single high-school snapshot** (baseline tests). Outcome is **attainment many years later**. You are not watching within-year competition; you are asking whether **early peer composition** lines up with **later degree completion**.

3. Map to our HERO / Big Fish frame

Binding rule: **environment ≠ advancement**; **score ≠ select**. Education stress-tests the **sorting** and **peer context** legs more than **scarce top-K selection**.

Our term	NELS / HSB column(s)	Note
Unit / student	<code>student_id</code>	One person
Pond	<code>school_id</code> , <code>unit_id</code>	School (NELS: 10th-grade high school; HSB: school x Soph/Senior wave)
Own ability (Â)	<code>own_performance_raw</code> , <code>own_performance_z</code>	Baseline test composite (reading + history standardized scores)
Peer context (LOO)	<code>peer_mean_loo</code> , <code>peer_mean_z_loo</code>	Mean peer test score / z, excl. self (~97–100% fill)
Team mean ($\bar{T}_j$ analog)	<code>unit_mean_including_self</code>	School mean including self
Relative standing	<code>relative_performance_z</code> , <code>within_unit_percentile_loo</code>	Own vs peers
Congestion-ish proxies	<code>n_peers_above</code> , <code>share_peers_above</code> , <code>n_top_decile_peers_loo</code> , <code>gap_to_unit_top</code>	“How many strong peers in my sampled pond?” — not sports usage
Y (outcome)	NELS: <code>bachelors_or_higher_by_2000</code> · HSB: <code>bachelors_or_higher_by_1986</code>	Binary degree attainment , not draft/tenure
Sample gates	<code>analytic_sample_min5</code> , <code>analytic_sample_min10</code> , <code>cell_at_least_*</code>	Minimum peers for stable LOO
Survey weights	NELS: <code>followup_weight_2000</code> · HSB: <code>followup_weight_1986</code> , <code>base_year_weight</code>	Use for population-style rates in serious tables

What is missing vs sports panels: No Alex `own_performance_index` composite, no season-by-season panel, no explicit **top-K winner rule** in the data.

4. Key numbers (analytic samples)

Panel	Filter	N	P(BA or higher)	Median peers
NELS:88	<code>analytic_sample_min5 == 1</code>	7,238	~36%	~13 peers
HS&B Sophomore	<code>analytic_sample_min5 == 1</code>	12,184	~6.7%	(see CSV)
HS&B Senior	<code>analytic_sample_min5 == 1</code>	9,874	~17%	(see CSV)

HSB rule (Alex): Analyze **Sophomore and Senior separately** — different samples and questionnaires; **do not pool** into one HERO without justification.

Why HSB BA rates differ: 1986 follow-up is **early** for a BA (many still in college). NELS 2000 is a more mature attainment window.

Other Y columns (secondary): `associate_or_higher`, `any_postsecondary`, `high_school_diploma` — useful for robustness, not primary dissertation porch.

5. How to interpret — three Alex asks

Alex’s Sep 2026 guidance: screen for **(1) congestion**, **(2) assortativity**, **(3) effect** (does peer context move the outcome porch?).

5.1 Assortativity (sorting index H_{sort})

Question: Are students **sorted into schools** by baseline test ability?

- Compute H_{sort} on `own_performance_z` with pools = **school** (same recipe as MBB overlap / Army — `541_grandchild_homophily_assign.realized_sorting_index_H_sort`).
- **Expectation:** Education should show **higher H_{sort}** than MBB PPM (~0.06) and plausibly in the **tenure (~0.17) or Army (~0.27)** ballpark — school quality and tracking sort peers.
- **Panel:** Team/school **interval overlap** (same machinery as Army panel 5).

Read: High H_{sort} means **ASSIGN / homophily matters for pool formation** (“why am I in this school?”). It does **not** by itself prove congestion-in-score (λ).

5.2 Peer context / "effect" (HERO-style porch)

Question: Does peer LOO predict P(BA+)?

- X: `peer_mean_z_loo` (or quantile bins thereof).
- Y: `bachelors_or_higher_by_*`.
- Plot: P(BA+) by peer LOO ventile — same Layer A grammar as Army/MBB HERO.

Literature baseline: "Big fish in a small pond" — often **monotonic** (weaker peers → higher attainment or self-concept), not necessarily MBB-style **inverted-U**.

Do not over-claim: This is **attainment**, not **one scarce slot per school**. ~36% BA in NELS is a **population rate**, not NBA draft odds.

5.3 Congestion (sports sense)

Question: Is there a **scarce-resource** / **spotlight** story?

- **Weak direct analog** in education vs football **usage LOO**.
- Closest panel columns: `share_top_decile_peers_loo`, `n_peers_above`, `gap_to_unit_top` — "how crowded is the top of my sampled pond?"
- **Pass A** (λ knockout) is **not in the CSV** — education does not observe a ranking equation.

Honest framing: Education is strongest on **sorting + peer composition** → **long-run outcome**, not on **congestion in a score that ranks candidates for one slot**.

6. What to claim vs not claim

OK to say	Avoid
Schools sort students by measured ability ($H_{\text{sort}} > 0$).	"Education confirms MBB-style inverted-U hero" without seeing the curve.
Peer LOO at baseline predicts (or doesn't) BA+ many years later.	"Peer LOO causes degree completion."
Education extends the framework to public, non-sports data.	"Same selection mechanism as NFL draft / tenure vote."
Sampled-classmate LOO is explicitly partial peers.	"Full school roster LOO."
HSB Soph vs Senior are different cohorts .	One pooled "HSB HERO" without split.

7. Relation to homophily (ρ) and MBB "works anyway"

Three objects — keep separate:

Object	Role
H_{sort}	Empirical sorting — variance in ability explained by pond assignment
ρ (homophily)	Simulated ASSIGN knob calibrated to match H_{sort} (PD21)
λ (congestion in score)	Scoring in $S_i = A_i - \lambda L_C$ — Pass A knockout

2a — Can we get congestion effects without assortativity?

Yes in the sim: fixed rosters, toggle λ , same top-K select. Congestion-in-score does **not** require homophily.

2b — MBB boundary case: MBB has **very low** H_{sort} (~0.06) and **tiny** K/N (few draft slots per huge roster). Hero-ish curves may appear even when $\rho^* \approx 0$. Army ($H_{\text{sort}} \approx 0.27$) is the opposite regime — **real sorting into units**.

Education's role in that debate: Likely **high** H_{sort} , **non-top-K Y** — separates "**sorted ponds**" from "**scarce selective advancement**." Even a flat HERO porch would be informative: *assortativity without draft-like selection*.

8. Suggested screening sequence (when you build)

1. **NELS:88 first** (smaller, cleaner outcome window by 2000).
 - Overlap + H_{sort} on school partition.
 - HERO: P(BA+) vs `peer_mean_z_loo` (Q16 ventiles); use `followup_weight_2000` for weighted rates if reporting population-style numbers.
 - Gate: `analytic_sample_min10 == 1`.
2. **HS&B:80 — Soph and Senior as two decks**.
 - Same panels; note lower BA rate by 1986, especially Sophomores.
3. **One-page synthesis table** for Alex:

Domain	H_{sort}	HERO shape?	Congestion proxy?	Selection-like Y?
NELS	(fill in)			No (~36% BA)
HSB Soph				No (~7% BA)
HSB Senior				No (~17% BA)

Not in this primer but same portfolio: Football (position + usage congestion), LoL (dev→top-tier promotion). See `HEROs_and_PASSes/_DISPOSABLE_big_fish_datasets_assessment.md`.

9. Column cheat sheet (both panels)

Shared Big Fish columns (names align across NELs and HSB):

Column	Meaning
<code>own_performance_raw</code>	Baseline test composite (level)
<code>own_performance_z</code>	Student z-score (global or as documented in Alex build)
<code>peer_mean_loo</code>	Mean peer raw score, excl. self
<code>peer_mean_z_loo</code>	Mean peer z, excl. self — primary HERO X
<code>peer_sd_loo</code>	Dispersion of peers
<code>unit_n</code>	Sample size in school cell (incl. self)
<code>peer_n</code>	Peers used for LOO (= unit_n – 1)
<code>within_unit_percentile_loo</code>	Percentile within school sample
<code>n_peers_above</code> / <code>share_peers_above</code>	Count / share of peers with higher raw score
<code>n_top_decile_peers_loo</code>	Count of global top-decile peers in pond
<code>unit_mean_including_self</code>	School mean including self
<code>gap_to_unit_top</code>	Distance to top scorer in cell
<code>global_top_decile_cutoff</code> / <code>own_global_top_decile</code>	Global high-ability flags
<code>base_year_ses</code>	SES control
<code>sex_label</code> / <code>race_label</code> (NELs) or <code>sex</code> / <code>race</code> (HSB)	Demographics

HSB-only extras: `gpa10`, `gpa12`, `school_type_label`, `school_region`, `SOQFLAG` / `SRQFLAG` (questionnaire flags).

10. One-sentence talk tracks

For Alex (education value):
“NELs and HS&B let us show **sorted school ponds** and **peer LOO vs long-run attainment** on public data — high assortativity without pretending every domain is NFL draft selection.”

For the dissertation boundary:
“Education is our **assortativity** and **peer-composition** leg; Army/MBB/tenure remain the **scarce advancement** legs.”

For methods footnote:
“Peer LOO is over **sampled classmates**, not full rosters; weights matter for population rates.”

11. Related repo docs

Doc	Role
<code>HEROs_and_PASSes/_DISPOSABLE_big_fish_datasets_assessment.md</code>	Portfolio verdict + sequencing
<code>3-Master_Plan/BINDING_Selection_is_its_own_step.md</code>	Environment vs advancement; score vs select
<code>HEROs_and_PASSes/legends_sandbox/LEGENDS_LoL_primer_for_researchers.md</code>	Parallel primer format (sports domain)
<code>scripts/big_fish_data_story.py</code>	Football + LoL mosaics today — education not wired yet

End of primer.